

A Rapid Multiplex Assay for Pathogen Detection

Abstract

This paper presents a multiplex rapid assay that combines target amplification, spatial encoding, and fluorescence readout for low-resource pathogen detection. The method aims to reduce workflow complexity when symptoms overlap.

Introduction

Single-target assays are inefficient for co-infection scenarios and require repeated testing.

Materials and Methods

The authors designed primer sets, mapped targets to spatial codes, captured fluorescence, and used threshold rules to call positive reactions.

Results

The assay distinguished multiple targets in a single workflow. The encoded readout reduced interpretation complexity and worked on simulated samples.

Figure 1. Overall workflow from sample input to encoded fluorescence readout.

Figure 2. Primer screening and target design strategy.

Discussion

The sample size is limited and clinical generalization requires further validation.